

Predicting PID5 scores from tract metrics in the group without PTSD

Loading FD data...

FD → Negative Affectivity

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1459 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.971, p=0.178 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Negative Affectivity

formula: Negative_Affectivity_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable: Negative_Affectivity_z No. Observations: 57
Model: RLM Df Residuals: 52
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 13:53:58
No. Iterations: 13

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0291	0.130	-0.224	0.823	-0.284	0.225
UF_fd_mean_z	-0.3526	0.189	-1.863	0.062	-0.724	0.018
CB_fd_mean_z	0.2674	0.175	1.531	0.126	-0.075	0.610
ST_fd_mean_z	0.1538	0.168	0.916	0.360	-0.175	0.483
mri_age_z	-0.3027	0.183	-1.654	0.098	-0.662	0.056

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FD → Detachment

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4695 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.949, p=0.018 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Detachment

formula: Detachment_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable: Detachment_z No. Observations: 57
Model: RLM Df Residuals: 52
Method: IRLS Df Model: 4
Norm: HuberT
Scale Est.: mad
Cov Type: H1
Date: Mon, 21 Sep 2026
Time: 13:53:58
No. Iterations: 11

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0325	0.134	-0.243	0.808	-0.294	0.229
UF_fd_mean_z	-0.0247	0.195	-0.127	0.899	-0.407	0.357

CB_fd_mean_z	0.0229	0.180	0.128	0.898	-0.329	0.375
ST_fd_mean_z	-0.1727	0.173	-0.999	0.318	-0.511	0.166
mri_age_z	-0.0889	0.188	-0.472	0.637	-0.458	0.280

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD -> Antagonism
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.0820 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.957, p=0.041 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD -> Antagonism

formula: Antagonism_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

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Robust linear Model Regression Results
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Dep. Variable:    Antagonism_z  No. Observations:      57
Model:            RLM  Df Residuals:      52
Method:           IRLS  Df Model:         4
Norm:             HuberT
Scale Est.:      mad
Cov Type:        H1
Date:            Mon, 21 Sep 2026
Time:            13:53:59
No. Iterations:   12
=====
              coef  std err          z      P>|z|    [0.025    0.975]
-----
Intercept  -0.0488    0.127    -0.384    0.701    -0.297    0.200
UF_fd_mean_z  -0.0119    0.185    -0.064    0.949    -0.374    0.351
CB_fd_mean_z   0.0017    0.171     0.010    0.992    -0.333    0.336
ST_fd_mean_z  -0.0210    0.164    -0.128    0.898    -0.343    0.301
mri_age_z    -0.3383    0.179    -1.891    0.059    -0.689    0.012
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If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD -> Disinhibition
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1514 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.968, p=0.143 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD -> Disinhibition

formula: Disinhibition_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

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=====
Robust linear Model Regression Results
=====
Dep. Variable:    Disinhibition_z  No. Observations:      57
Model:            RLM  Df Residuals:      52
Method:           IRLS  Df Model:         4
Norm:             HuberT
Scale Est.:      mad
Cov Type:        H1
Date:            Mon, 21 Sep 2026
Time:            13:53:59
No. Iterations:   19
=====
              coef  std err          z      P>|z|    [0.025    0.975]
-----
Intercept  -0.0485    0.136    -0.356    0.722    -0.315    0.218
UF_fd_mean_z  -0.1053    0.198    -0.531    0.595    -0.494    0.283
CB_fd_mean_z   0.1561    0.183     0.854    0.393    -0.202    0.514
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ST_fd_mean_z -0.3080 0.176 -1.752 0.080 -0.653 0.037
mri_age_z -0.3788 0.192 -1.976 0.048 -0.754 -0.003
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```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Psychoticism
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1009 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.980, p=0.453 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Psychoticism

formula: Psychoticism_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results							
=====							
Dep. Variable:	Psychoticism_z	No. Observations:	57				
Model:	RLM	Df Residuals:	52				
Method:	IRLS	Df Model:	4				
Norm:	HuberT						
Scale Est.:	mad						
Cov Type:	H1						
Date:	Mon, 21 Sep 2026						
Time:	13:53:59						
No. Iterations:	19						
=====							
	coef	std err	z	P> z	[0.025	0.975]	

Intercept	-0.0049	0.141	-0.035	0.972	-0.280	0.270	
UF_fd_mean_z	-0.0946	0.205	-0.462	0.644	-0.496	0.307	
CB_fd_mean_z	0.0754	0.189	0.399	0.690	-0.295	0.446	
ST_fd_mean_z	-0.1620	0.182	-0.892	0.372	-0.518	0.194	
mri_age_z	-0.4750	0.198	-2.398	0.016	-0.863	-0.087	

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

Loading FC data...

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FC → Negative Affectivity
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.9742 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.983, p=0.616 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Negative Affectivity

formula: Negative_Affectivity_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results			
Dep. Variable:	Negative_Affectivity_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	13:53:59		
No. Iterations:	14		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0084	0.148	-0.057	0.955	-0.299	0.282
UF_fc_mean_res_z	-0.1463	0.157	-0.933	0.351	-0.454	0.161
CB_fc_mean_res_z	0.1425	0.169	0.842	0.400	-0.189	0.474
ST_fc_mean_res_z	0.0359	0.180	0.200	0.842	-0.317	0.388
mri_age_z	-0.3029	0.182	-1.660	0.097	-0.660	0.055

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Detachment

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4888 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.964, p=0.089 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Detachment

formula: Detachment_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Detachment_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	13:54:00		
No. Iterations:	15		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0402	0.129	-0.311	0.756	-0.294	0.213
UF_fc_mean_res_z	-0.2539	0.137	-1.853	0.064	-0.522	0.015
CB_fc_mean_res_z	0.0060	0.148	0.041	0.967	-0.283	0.295
ST_fc_mean_res_z	0.1086	0.157	0.691	0.489	-0.199	0.416
mri_age_z	-0.0330	0.159	-0.208	0.836	-0.345	0.279

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Antagonism

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1714 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.954, p=0.029 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Antagonism

formula: Antagonism_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Antagonism_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	13:54:00		
No. Iterations:	20		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0408	0.115	-0.354	0.723	-0.267	0.185
UF_fc_mean_res_z	-0.2643	0.122	-2.164	0.030	-0.504	-0.025
CB_fc_mean_res_z	0.4501	0.132	3.419	0.001	0.192	0.708
ST_fc_mean_res_z	0.1074	0.140	0.767	0.443	-0.167	0.382
mri_age_z	-0.5122	0.142	-3.609	0.000	-0.790	-0.234

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Disinhibition

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6156 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.958, p=0.046 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Disinhibition

formula: Disinhibition_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Disinhibition_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	13:54:00		
No. Iterations:	23		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0694	0.139	-0.499	0.618	-0.342	0.203
UF_fc_mean_res_z	-0.0978	0.147	-0.664	0.507	-0.387	0.191
CB_fc_mean_res_z	0.2044	0.159	1.288	0.198	-0.107	0.516
ST_fc_mean_res_z	0.1995	0.169	1.182	0.237	-0.131	0.530
mri_age_z	-0.3127	0.171	-1.827	0.068	-0.648	0.023

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Psychoticism

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.1169 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.982, p=0.559 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Psychoticism

formula: Psychoticism_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	Psychoticism_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	13:54:00		
No. Iterations:	16		

	coef	std err	z	P> z	[0.025	0.975]
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Intercept	-0.0056	0.144	-0.039	0.969	-0.288	0.276
UF_fc_mean_res_z	-0.0955	0.152	-0.627	0.531	-0.394	0.203
CB_fc_mean_res_z	0.0421	0.164	0.256	0.798	-0.280	0.364
ST_fc_mean_res_z	0.1625	0.175	0.930	0.352	-0.180	0.505
mri_age_z	-0.4424	0.177	-2.496	0.013	-0.790	-0.095

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

All models complete.

Predicting CTQ scores from tract metrics in the group without PTSD

Loading FD data...

FD → Total CTQ

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6897 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.982, p=0.531 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Total CTQ

formula: CTQ_total_score_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQ_total_score_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:27		
No. Iterations:	20		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0202	0.148	-0.137	0.891	-0.310	0.269
UF_fd_mean_z	0.2223	0.215	1.033	0.301	-0.199	0.644
CB_fd_mean_z	-0.1796	0.198	-0.905	0.366	-0.569	0.209
ST_fd_mean_z	-0.0455	0.191	-0.239	0.811	-0.420	0.328
mri_age_z	0.2527	0.208	1.215	0.225	-0.155	0.661

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FD → Emotional Abuse

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.8699 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.962, p=0.073 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Emotional Abuse

formula: CTQemotional_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQemotional_abuse_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:27		
No. Iterations:	19		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0333	0.138	-0.242	0.809	-0.304	0.237
UF_fd_mean_z	0.0697	0.201	0.347	0.729	-0.324	0.464

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CB_fd_mean_z -0.1941 0.186 -1.046 0.295 -0.558 0.170
ST_fd_mean_z -0.0855 0.178 -0.479 0.632 -0.435 0.264
mri_age_z 0.0422 0.194 0.217 0.828 -0.339 0.423
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```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Physical Abuse
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Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.8645 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.902, p=0.000 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Physical Abuse

formula: CTQphysical_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable:  CTQphysical_abuse_z  No. Observations:      57
Model:          RLM  Df Residuals:          52
Method:         IRLS  Df Model:             4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:28
No. Iterations: 17
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0816  0.121  -0.674  0.501  -0.319  0.156
UF_fd_mean_z  0.0151  0.177   0.086  0.932  -0.331  0.361
CB_fd_mean_z -0.2631  0.163  -1.615  0.106  -0.582  0.056
ST_fd_mean_z -0.0570  0.157  -0.364  0.716  -0.364  0.250
mri_age_z   -0.1018  0.171  -0.596  0.551  -0.436  0.233
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

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FD → Sexual Abuse
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```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.2208 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.591, p=0.000 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Sexual Abuse

formula: CTQsexual_abuse_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

```
Robust linear Model Regression Results
=====
Dep. Variable:  CTQsexual_abuse_z  No. Observations:      57
Model:          RLM  Df Residuals:          52
Method:         IRLS  Df Model:             4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:28
No. Iterations: 50
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.2781  0.052  -5.316  0.000  -0.381 -0.176
UF_fd_mean_z -0.0274  0.076  -0.359  0.720  -0.177  0.122
CB_fd_mean_z  0.0084  0.070   0.120  0.905  -0.129  0.146
=====
```

```
ST_fd_mean_z -0.0872 0.068 -1.289 0.198 -0.220 0.045
mri_age_z -0.0753 0.074 -1.020 0.308 -0.220 0.069
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Emotional Neglect
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.6472 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.980, p=0.459 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Emotional Neglect

formula: CTQemotional_neglect_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

```
=====
Dep. Variable:  CTQemotional_neglect_z  No. Observations:  57
Model:          RLM  Df Residuals:  52
Method:         IRLS  Df Model:  4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:28
No. Iterations:  23
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0304  0.151  -0.201  0.841  -0.327  0.266
UF_fd_mean_z  0.1893  0.221  0.858  0.391  -0.243  0.622
CB_fd_mean_z  0.0060  0.204  0.029  0.977  -0.393  0.405
ST_fd_mean_z -0.0123  0.196  -0.063  0.950  -0.396  0.371
mri_age_z    0.3897  0.213  1.826  0.068  -0.028  0.808
=====
```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```
=====
FD → Physical Neglect
=====
```

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.9236 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.934, p=0.004 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fd_mean_z 2.086994
CB_fd_mean_z 1.776690
ST_fd_mean_z 1.642043
mri_age_z 1.952289
Intercept 1.000000

- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FD → Physical Neglect

formula: CTQphys_neglect_z ~ UF_fd_mean_z + CB_fd_mean_z + ST_fd_mean_z + mri_age_z

Robust linear Model Regression Results

```
=====
Dep. Variable:  CTQphys_neglect_z  No. Observations:  57
Model:          RLM  Df Residuals:  52
Method:         IRLS  Df Model:  4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:28
No. Iterations:  15
=====
      coef  std err      z  P>|z|  [0.025  0.975]
-----
Intercept -0.0877  0.123  -0.715  0.474  -0.328  0.153
UF_fd_mean_z  0.1109  0.179  0.621  0.535  -0.239  0.461
CB_fd_mean_z -0.0341  0.165  -0.207  0.836  -0.357  0.289
ST_fd_mean_z  0.2538  0.158  1.602  0.109  -0.057  0.564
=====
```

```
mri_age_z    0.2326    0.173    1.346    0.178    -0.106    0.571
```

=====

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

=====

Loading FC data...

=====

FC → Total CTQ

=====

Assumption checks

=====

- [1] Homoscedasticity
Breusch-Pagan p=0.2687 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.987, p=0.816 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Total CTQ

=====

formula: CTQ_total_score_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

=====

Dep. Variable:	CTQ_total_score_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:29		
No. Iterations:	18		

=====

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0237	0.153	-0.155	0.877	-0.323	0.276
UF_fc_mean_res_z	-0.0221	0.162	-0.137	0.891	-0.339	0.295
CB_fc_mean_res_z	-0.0798	0.175	-0.457	0.648	-0.422	0.262
ST_fc_mean_res_z	-0.0165	0.186	-0.089	0.929	-0.380	0.347
mri_age_z	0.2097	0.188	1.114	0.265	-0.159	0.579

=====

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

=====

FC → Emotional Abuse

=====

Assumption checks

=====

- [1] Homoscedasticity
Breusch-Pagan p=0.9244 — homoscedasticity assumed.
- [2] Normality of residuals
Shapiro-Wilk W=0.971, p=0.187 — normality of residuals assumed.
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Emotional Abuse

=====

formula: CTQemotional_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

=====

Dep. Variable:	CTQemotional_abuse_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:29		
No. Iterations:	16		

=====

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0298	0.144	-0.206	0.837	-0.313	0.253
UF_fc_mean_res_z	-0.1319	0.153	-0.862	0.389	-0.432	0.168
CB_fc_mean_res_z	-0.0742	0.165	-0.449	0.653	-0.398	0.249
ST_fc_mean_res_z	0.0087	0.175	0.050	0.960	-0.335	0.353
mri_age_z	0.0930	0.178	0.523	0.601	-0.256	0.442

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Physical Abuse

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.7862 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.862, p=0.000 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Physical Abuse

formula: CTQphysical_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQphysical_abuse_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:29		
No. Iterations:	41		

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0822	0.130	-0.634	0.526	-0.336	0.172
UF_fc_mean_res_z	0.0053	0.137	0.038	0.969	-0.264	0.275
CB_fc_mean_res_z	-0.0974	0.148	-0.658	0.511	-0.388	0.193
ST_fc_mean_res_z	-0.0398	0.158	-0.253	0.800	-0.349	0.269
mri_age_z	-0.0057	0.160	-0.035	0.972	-0.319	0.307

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

FC → Sexual Abuse

Assumption checks

- [1] Homoscedasticity
Breusch-Pagan p=0.4540 — homoscedasticity assumed.
- [2] Normality of residuals
WARNING: Shapiro-Wilk W=0.588, p=0.000 — residuals not normally distributed! (Not a formal requirement for Huber regression)
- [3] Multicollinearity (VIF)
No problematic multicollinearity detected.
Variable VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z 1.490573
Intercept 1.000000
- [4] Linearity
Inspect plot: residuals should form a random cloud around zero.

MLR results: FC → Sexual Abuse

formula: CTQsexual_abuse_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

Robust linear Model Regression Results

Dep. Variable:	CTQsexual_abuse_z	No. Observations:	57
Model:	RLM	Df Residuals:	52
Method:	IRLS	Df Model:	4
Norm:	HuberT		
Scale Est.:	mad		
Cov Type:	H1		
Date:	Mon, 21 Sep 2026		
Time:	14:02:30		
No. Iterations:	50		

	coef	std err	z	P> z	[0.025	0.975]
--	------	---------	---	------	--------	--------

```

-----
Intercept      -0.2948   0.048   -6.159   0.000   -0.389   -0.201
UF_fc_mean_res_z  0.0376   0.051   0.742   0.458   -0.062   0.137
CB_fc_mean_res_z -0.0250   0.055  -0.457   0.648   -0.132   0.082
ST_fc_mean_res_z  0.0524   0.058   0.902   0.367   -0.062   0.166
mri_age_z       -0.0263   0.059  -0.446   0.656   -0.142   0.089
-----

```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```

=====
FC → Emotional Neglect
=====

```

Assumption checks

```

-----
[1] Homoscedasticity
    Breusch-Pagan p=0.9796 — homoscedasticity assumed.

[2] Normality of residuals
    Shapiro-Wilk W=0.982, p=0.554 — normality of residuals assumed.
/home/amygdala/miniconda3/envs/project/lib/python3.11/site-packages/statsmodels/graphics/utils.py:55: RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot
interface ('matplotlib.pyplot.figure') are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam 'figure.max_open_warning'). Consider using
'matplotlib.pyplot.close()'.
  fig = plt.figure()

[3] Multicollinearity (VIF)
    No problematic multicollinearity detected.
    Variable    VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z        1.490573
Intercept        1.000000

[4] Linearity
    Inspect plot: residuals should form a random cloud around zero.
-----

```

MLR results: FC → Emotional Neglect

formula: CTQemotional_neglect_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

```

=====
Robust linear Model Regression Results
=====
Dep. Variable:  CTQemotional_neglect_z  No. Observations:      57
Model:          RLM  Df Residuals:      52
Method:         IRLS  Df Model:         4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:30
No. Iterations: 19
=====
coef  std err      z  P>|z|  [0.025   0.975]
-----
Intercept  -0.0176   0.152  -0.116   0.908   -0.316   0.280
UF_fc_mean_res_z -0.0136   0.161  -0.084   0.933   -0.329   0.302
CB_fc_mean_res_z -0.0181   0.174  -0.104   0.917   -0.359   0.322
ST_fc_mean_res_z  0.0356   0.185   0.193   0.847   -0.326   0.398
mri_age_z     0.2695   0.187   1.439   0.150   -0.098   0.637
=====

```

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

```

=====
FC → Physical Neglect
=====

```

Assumption checks

```

-----
[1] Homoscedasticity
    Breusch-Pagan p=0.7502 — homoscedasticity assumed.

[2] Normality of residuals
    WARNING: Shapiro-Wilk W=0.939, p=0.006 — residuals not normally distributed! (Not a formal requirement for Huber regression)

[3] Multicollinearity (VIF)
    No problematic multicollinearity detected.
    Variable    VIF
UF_fc_mean_res_z 1.103108
CB_fc_mean_res_z 1.282040
ST_fc_mean_res_z 1.449581
mri_age_z        1.490573
Intercept        1.000000

[4] Linearity
    Inspect plot: residuals should form a random cloud around zero.
-----

```

MLR results: FC → Physical Neglect

formula: CTQphys_neglect_z ~ UF_fc_mean_res_z + CB_fc_mean_res_z + ST_fc_mean_res_z + mri_age_z

```

=====
Robust linear Model Regression Results
=====
Dep. Variable:  CTQphys_neglect_z  No. Observations:      57
Model:          RLM  Df Residuals:      52
Method:         IRLS  Df Model:         4
Norm:           HuberT
Scale Est.:     mad
Cov Type:       H1
Date:           Mon, 21 Sep 2026
Time:           14:02:30

```

No. Iterations: 15

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-0.0895	0.123	-0.725	0.469	-0.331	0.153
UF_fc_mean_res_z	0.0659	0.131	0.504	0.614	-0.191	0.322
CB_fc_mean_res_z	-0.0761	0.141	-0.540	0.589	-0.353	0.200
ST_fc_mean_res_z	-0.2500	0.150	-1.667	0.095	-0.544	0.044
mri_age_z	0.1935	0.152	1.272	0.203	-0.105	0.492

If the model instance has been used for another fit with different fit parameters, then the fit options might not be the correct ones anymore .

All models complete.